

Iteration : 1

 $a[0] > a[1]$

[0]	[1]	[2]	[3]	[4]	[5]
30	20	60	50	10	40

Sel_pos pos

30 > 20 ? Yes

swap

 $a[0] > a[2]$

[0]	[1]	[2]	[3]	[4]	[5]
20	30	60	50	10	40

Sel_pos pos

20 > 60 ? no

 $a[0] > a[3]$

[0]	[1]	[2]	[3]	[4]	[5]
20	30	60	50	10	40

Sel_pos pos

20 > 50 ? no

 $a[0] > a[4]$

[0]	[1]	[2]	[3]	[4]	[5]
20	30	60	50	10	40

Sel_pos pos

20 > 10 ? Yes

swap

 $a[0] > a[5]$

[0]	[1]	[2]	[3]	[4]	[5]
10	30	60	50	20	40

Sel_pos pos

10 > 40 ? no

[0]	[1]	[2]	[3]	[4]	[5]
10	30	60	50	20	40

Sel_pos

Iteration /
cycle/ pass : 2

[0]	[1]	[2]	[3]	[4]	[5]
10	30	60	50	20	40

Sel_pos pos

30 > 60 ? no

[0]	[1]	[2]	[3]	[4]	[5]
10	30	60	50	20	40

Sel_pos pos

30 > 50 ? no

[0]	[1]	[2]	[3]	[4]	[5]
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Sel_pos pos

[0]	[1]	[2]	[3]	[4]	[5]
10	30	60	50	20	40

30 > 20 ? Yes
swap

Sel_pos pos

[0]	[1]	[2]	[3]	[4]	[5]
10	20	60	50	30	40

20 > 40 ? no

Sel_pos pos

[0]	[1]	[2]	[3]	[4]	[5]
10	20	60	50	30	40

Iteration : 3

[0]	[1]	[2]	[3]	[4]	[5]
10	20	60	50	30	40

60 > 50 ? Yes
swap

Sel_pos pos

[0]	[1]	[2]	[3]	[4]	[5]
10	20	50	60	30	40

50 > 30 ? Yes
swap

Sel_pos pos

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	60	50	40

30 > 40 ? no

Sel_pos pos

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	60	50	40

Iteration : 4

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	60	50	40

60 > 50 ? Yes
Arr[3] > arr[4]

Sel_pos pos

Arr[sel_pos] > arr[pos] ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	50	60	40

Arr[sel_pos] > arr[pos] ? Yes
swap

Sel_pos pos

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	60	50

Iteration : 5

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	60	50

Sel_pos pos

60 > 50 ? Yes
swap

[0]	[1]	[2]	[3]	[4]	[5]
10	20	30	40	50	60

No. of iterations : $n-1$: 5

No. of Comparisons : $(n-1) + (n-2) + (n-3) \dots 1$
 $5 + 4 + 3 + 2 + 1 = 15$